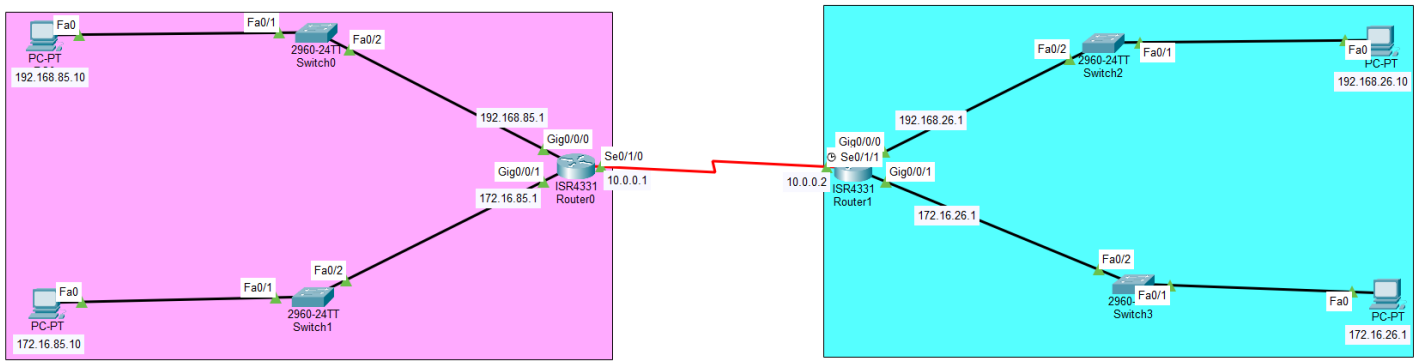
1. Network topology



2. IP addressing table

Device	Interface	IP Address	Subnet Mask	Wildcard / Prefix	Description / Purpose
R1_Anup2607985	Gig0/0/0	192.168.85.1	255.255.255.0	/24	Gateway for Site A - Switch0 LAN
R1_Anup2607985	Gig0/0/1	172.16.85.1	255.255.255.0	/24	Gateway for Site A - Switch1 LAN
R1_Anup2607985	Ser0/1/0	10.0.0.1	255.255.255.252	/30	WAN Interface to R2
DHCP Pool: SITE_A_SW0	192.168.85.0	Gateway: 192.168.85.1	Range: .10 - .254	/24	Dynamic IP range for PC0
DHCP Pool: SITE_A_SW1	172.16.85.0	Gateway: 172.16.85.1	Range: .10 - .254	/24	Dynamic IP range for PC1

Device	Interface	IP Address	Subnet Mask	Wildcard / Prefix	Description / Purpose
R2_Anup2607985	Gig0/0/0	192.168.26.1	255.255.255.0	/24	Gateway for Site B - Switch2 LAN
R2_Anup2607985	Gig0/0/1	172.16.26.1	255.255.255.0	/24	Gateway for Site B - Switch3 LAN
R2_Anup2607985	Ser0/1/1	10.0.0.2	255.255.255.252	/30	WAN Interface to R1
DHCP Pool: SITE_B_SW2	192.168.26.0	Gateway: 192.168.26.1	Range: .10 - .254	/24	Dynamic IP range for PC2
DHCP Pool: SITE_B_SW3	172.16.26.0	Gateway: 172.16.26.1	Range: .10 - .254	/24	Dynamic IP range for PC3

VPN Testing and Verification

```
R1_Anup2607985#show crypto isakmp policy

Global IKE policy
Protection suite of priority 10
        encryption algorithm:   AES - Advanced Encryption Standard (128 bit keys).
        hash algorithm:         Secure Hash Standard
        authentication method:  Pre-Shared Key
        Diffie-Hellman group:   #2 (1024 bit)
        lifetime:               86400 seconds, no volume limit
R1_Anup2607985#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst             src             state          conn-id slot status
10.0.0.2        10.0.0.1        QM_IDLE           1020    0 ACTIVE


IPv6 Crypto ISAKMP SA


R1_Anup2607985#show crypto ipsec sa

interface: Serial0/1/0
    Crypto map tag: CMAP-VPN, local addr 10.0.0.1

   protected vrf: (none)
   local  ident (addr/mask/prot/port): (192.168.85.0/255.255.255.0/0/0)
   remote  ident (addr/mask/prot/port): (192.168.26.0/255.255.255.0/0/0)
   current_peer 10.0.0.2 port 500
    PERMIT, flags={origin_is_acl,}
   #pkts encaps: 0, #pkts encrypt: 0, #pkts digest: 0
   #pkts decaps: 0, #pkts decrypt: 0, #pkts verify: 0
   #pkts compressed: 0, #pkts decompressed: 0
   #pkts not compressed: 0, #pkts compr. failed: 0
   #pkts not decompressed: 0, #pkts decompress failed: 0
   #send errors 0, #recv errors 0

     local crypto endpt.: 10.0.0.1, remote crypto endpt.:10.0.0.2
     path mtu 1500, ip mtu 1500, ip mtu idb Serial0/1/0
     current outbound spi: 0x0(0)

     inbound esp sas:
```

```
R1_Anup2607985#show crypto map
Crypto Map CMAP-VPN 10 ipsec-isakmp
        Peer = 10.0.0.2
        Extended IP access list VPN-ACL
            access-list VPN-ACL permit ip 192.168.85.0 0.0.0.255 192.168.26.0 0.0.0.255
            access-list VPN-ACL permit ip 192.168.85.0 0.0.0.255 172.16.26.0 0.0.0.255
            access-list VPN-ACL permit ip 172.16.85.0 0.0.0.255 192.168.26.0 0.0.0.255
            access-list VPN-ACL permit ip 172.16.85.0 0.0.0.255 172.16.26.0 0.0.0.255
        Current peer: 10.0.0.2
        Security association lifetime: 4608000 kilobytes/3600 seconds
        PFS (Y/N): N
        Transform sets={
                TS-VPN,
        }
        Interfaces using crypto map CMAP-VPN:
                Serial0/1/0
```

IPsec encryption/decryption evidence

```
R1_Anup2607985#show crypto ipsec sa

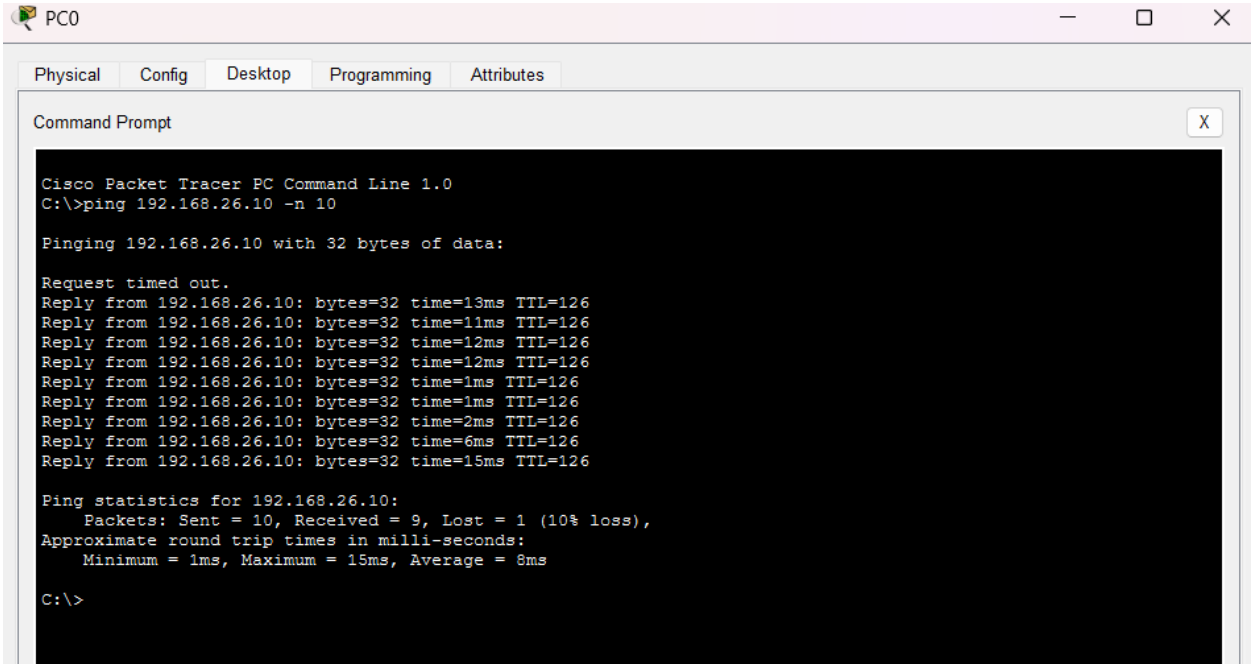
interface: Serial0/1/0
    Crypto map tag: CMAP-VPN, local addr 10.0.0.1

   protected vrf: (none)
   local  ident (addr/mask/prot/port): (192.168.85.0/255.255.255.0/0/0)
   remote  ident (addr/mask/prot/port): (192.168.26.0/255.255.255.0/0/0)
   current_peer 10.0.0.2 port 500
    PERMIT, flags={origin_is_acl,}
   #pkts encaps: 0, #pkts encrypt: 0, #pkts digest: 0
   #pkts decaps: 0, #pkts decrypt: 0, #pkts verify: 0
   #pkts compressed: 0, #pkts decompressed: 0
   #pkts not compressed: 0, #pkts compr. failed: 0
   #pkts not decompressed: 0, #pkts decompress failed: 0
   #send errors 0, #recv errors 0

     local crypto endpt.: 10.0.0.1, remote crypto endpt.:10.0.0.2
     path mtu 1500, ip mtu 1500, ip mtu idb Serial0/1/0
     current outbound spi: 0x0(0)

     inbound esp sas:
```

• Test connectivity between the PCs at the two sites and Generate traffic between the Site A and Site B networks



```
R1_Anup2607985#show crypto ipsec sa

interface: Serial0/1/0
    Crypto map tag: CMAP-VPN, local addr 10.0.0.1

   protected vrf: (none)
   local  ident (addr/mask/prot/port): (192.168.85.0/255.255.255.0/0/0)
   remote  ident (addr/mask/prot/port): (192.168.26.0/255.255.255.0/0/0)
   current_peer 10.0.0.2 port 500
    PERMIT, flags={origin_is_acl,}
   #pkts encaps: 9, #pkts encrypt: 9, #pkts digest: 0
   #pkts decaps: 9, #pkts decrypt: 9, #pkts verify: 0
   #pkts compressed: 0, #pkts decompressed: 0
   #pkts not compressed: 0, #pkts compr. failed: 0
   #pkts not decompressed: 0, #pkts decompress failed: 0
   #send errors 1, #recv errors 0

     local crypto endpt.: 10.0.0.1, remote crypto endpt.:10.0.0.2
     path mtu 1500, ip mtu 1500, ip mtu idb Serial0/1/0
     current outbound spi: 0xCF85DB86(3481656198)

     inbound esp sas:
      spi: 0xEDF7B9FA(3992435194)
        transform: esp-aes esp-sha-hmac ,
        in use settings ={Tunnel, }
        conn id: 2008, flow_id: FPGA:1, crypto map: CMAP-VPN
        sa timing: remaining key lifetime (k/sec): (4525504/3565)
        IV size: 16 bytes
        replay detection support: N
        Status: ACTIVE

     inbound ah sas:

     inbound pcp sas:

     outbound esp sas:
      spi: 0xCF85DB86(3481656198)
        transform: esp-aes esp-sha-hmac ,
        in use settings ={Tunnel, }
        conn id: 2009, flow_id: FPGA:1, crypto map: CMAP-VPN
        sa timing: remaining key lifetime (k/sec): (4525504/3565)
        IV size: 16 bytes
        replay detection support: N
        Status: ACTIVE

     outbound ah sas:

     outbound pcp sas:
```

```
local  ident (addr/mask/prot/port): (192.168.85.0/255.255.255.0/0/0)
remote  ident (addr/mask/prot/port): (172.16.26.0/255.255.255.0/0/0)
current_peer 10.0.0.2 port 500
 PERMIT, flags={origin_is_acl,}
#pkts encaps: 1, #pkts encrypt: 1, #pkts digest: 0
#pkts decaps: 1, #pkts decrypt: 1, #pkts verify: 0
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 1, #recv errors 0

  local crypto endpt.: 10.0.0.1, remote crypto endpt.:10.0.0.2
  path mtu 1500, ip mtu 1500, ip mtu idb Serial0/1/0
  current outbound spi: 0x48DF45C7(1222591943)

  inbound esp sas:
   spi: 0xDBB4D1DC(3686060508)
     transform: esp-aes esp-sha-hmac ,
     in use settings ={Tunnel, }
     conn id: 2003, flow_id: FPGA:1, crypto map: CMAP-VPN
     sa timing: remaining key lifetime (k/sec): (4525504/3036)
     IV size: 16 bytes
     replay detection support: N
     Status: ACTIVE

  inbound ah sas:

  inbound pcp sas:

  outbound esp sas:
   spi: 0x48DF45C7(1222591943)
     transform: esp-aes esp-sha-hmac ,
     in use settings ={Tunnel, }
     conn id: 2004, flow_id: FPGA:1, crypto map: CMAP-VPN
     sa timing: remaining key lifetime (k/sec): (4525504/3036)
     IV size: 16 bytes
     replay detection support: N
     Status: ACTIVE

  outbound ah sas:

  outbound pcp sas:
```

```
local  ident (addr/mask/prot/port): (172.16.85.0/255.255.255.0/0/0)
remote  ident (addr/mask/prot/port): (192.168.26.0/255.255.255.0/0/0)
current_peer 10.0.0.2 port 500
 PERMIT, flags={origin_is_acl,}
#pkts encaps: 1, #pkts encrypt: 1, #pkts digest: 0
#pkts decaps: 1, #pkts decrypt: 1, #pkts verify: 0
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 1, #recv errors 0

  local crypto endpt.: 10.0.0.1, remote crypto endpt.:10.0.0.2
  path mtu 1500, ip mtu 1500, ip mtu idb Serial0/1/0
  current outbound spi: 0x3611DF80(907140992)

  inbound esp sas:
   spi: 0x4E8123DB(1317086171)
     transform: esp-aes esp-sha-hmac ,
     in use settings ={Tunnel, }
     conn id: 2002, flow_id: FPGA:1, crypto map: CMAP-VPN
     sa timing: remaining key lifetime (k/sec): (4525504/3020)
     IV size: 16 bytes
     replay detection support: N
     Status: ACTIVE

  inbound ah sas:

  inbound pcp sas:

  outbound esp sas:
   spi: 0x3611DF80(907140992)
     transform: esp-aes esp-sha-hmac ,
     in use settings ={Tunnel, }
     conn id: 2003, flow_id: FPGA:1, crypto map: CMAP-VPN
     sa timing: remaining key lifetime (k/sec): (4525504/3020)
     IV size: 16 bytes
     replay detection support: N
     Status: ACTIVE

  outbound ah sas:

  outbound pcp sas:
```

```
local  ident (addr/mask/prot/port): (172.16.85.0/255.255.255.0/0/0)
remote  ident (addr/mask/prot/port): (172.16.26.0/255.255.255.0/0/0)
current_peer 10.0.0.2 port 500
 PERMIT, flags={origin_is_acl,}
#pkts encaps: 0, #pkts encrypt: 0, #pkts digest: 0
#pkts decaps: 0, #pkts decrypt: 0, #pkts verify: 0
#pkts compressed: 0, #pkts decompressed: 0
#pkts not compressed: 0, #pkts compr. failed: 0
#pkts not decompressed: 0, #pkts decompress failed: 0
#send errors 0, #recv errors 0

  local crypto endpt.: 10.0.0.1, remote crypto endpt.:10.0.0.2
  path mtu 1500, ip mtu 1500, ip mtu idb Serial0/1/0
  current outbound spi: 0x0(0)

  inbound esp sas:

  inbound ah sas:

  inbound pcp sas:

  outbound esp sas:

  outbound ah sas:

  outbound pcp sas:
```